Camille Billouard

PhD in AI
3D Computer Vision & Neural Rendering

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Google Scholar

Research interests

- Neural 3D reconstruction
- NeRF & neural rendering
- Scaling laws, large-scale ML
- Earth observation
- GPU / HPC efficiency

Education

2023–26
PhD, Artificial Intelligence
Université Gustave Eiffel
CNES / LASTIG (IGN)

2021–23
MSc, Artificial Intelligence
Université de Toulouse (IAFA)

2021
BSc, Computer Science
Université de Lille, research track

2020
BSc, Maths & Computer Science
Université de Bordeaux

Skills

ML / 3D
PyTorch, NeRF, nerfstudio, tiny-cuda-nn, nerfacc, volume rendering, multi-view & RPC geometry

Engineering
Python, CUDA, SLURM / HPC, Git, Java, SQL, Bash

Geospatial
xDEM, LiDAR HD, Pléiades Neo

Languages

French — native
English — fluent

Profile

AI researcher specialised in neural 3D reconstruction from satellite imagery at city scale. I turn research prototypes into fast, memory-bounded pipelines with PyTorch, custom CUDA and HPC, and I back engineering choices with theory and careful measurement.

Experience

Oct 2023 – Oct 2026 **PhD Researcher** | *CNES & LASTIG, Paris*

- Built **SAT-NGP**: satellite NeRF training **86–200× faster** (10–20 h down to 4–14 min), elevation error within 0.19 m of the state of the art; now a reference baseline for efficient 3D Earth observation.
- Built **Snake-NeRF**: out-of-core tiling that reconstructs **52 km²** of Pléiades Neo imagery (442 tiles) **on a single GPU**, with peak memory under 2 GB (within 1% of the analytical prediction), 5× lower host memory and near-linear runtime.
- Formalised scalability for neural rendering and derived a power-law scaling of reconstruction error with tile size.
- Wrote custom CUDA kernels (RPC ray generation) and DSM evaluation pipelines against IGN LiDAR HD on SLURM clusters.

Apr – Oct 2023 **Research Intern** | *CNES, Toulouse*
NeRF-based 3D reconstruction from multi-date satellite images.

May – Jul 2022 **Research Intern** | *IRIT (MINDS team), Toulouse*
Fast 3D super-resolution for small-field dental cone-beam CT volumes.

Apr – Aug 2021 **Software Engineering Intern** | *CGI, Bordeaux*
Java / Spring Batch jobs for banking process automation (Oracle SQL, GitLab, Sonar).

Publications

- **C. Billouard**, D. Derksen, A. Constantin, B. Vallet. *Tile and Slide: A New Framework for Scaling NeRF from Local to Global 3D Earth Observation.* ICCV Workshops, Hawaii, 2025. arXiv:2507.01631
- **C. Billouard**, D. Derksen, E. Sarrazin, B. Vallet. *SAT-NGP: Unleashing Neural Graphics Primitives for Fast Relightable Transient-Free 3D Reconstruction from Satellite Imagery.* IGARSS, Athens, 2024. doi
- K. Specogna, **C. Billouard**, et al. *Applications des champs de radiance neuronaux au mode vidéo de CO3D.* Revue Française de Photogrammétrie et de Télédétection (RFPT), 2026.

Talks & Seminars

Presented in person at **ICCV** (Hawaii, 2025), **IGARSS** (Athens, 2024), **ESA Living Planet Symposium**, **ORASIS**, **SFPT** and the **Meet'Up ML** workshop (Nantes).

Supervision

- Co-supervised two research interns: NeRF-based semantic segmentation (C. Herbin) and 3D reconstruction from CO3D video sequences (K. Specogna), published in RFPT 2026.